e-Power Technology

100 % Electric Motor-Driven system

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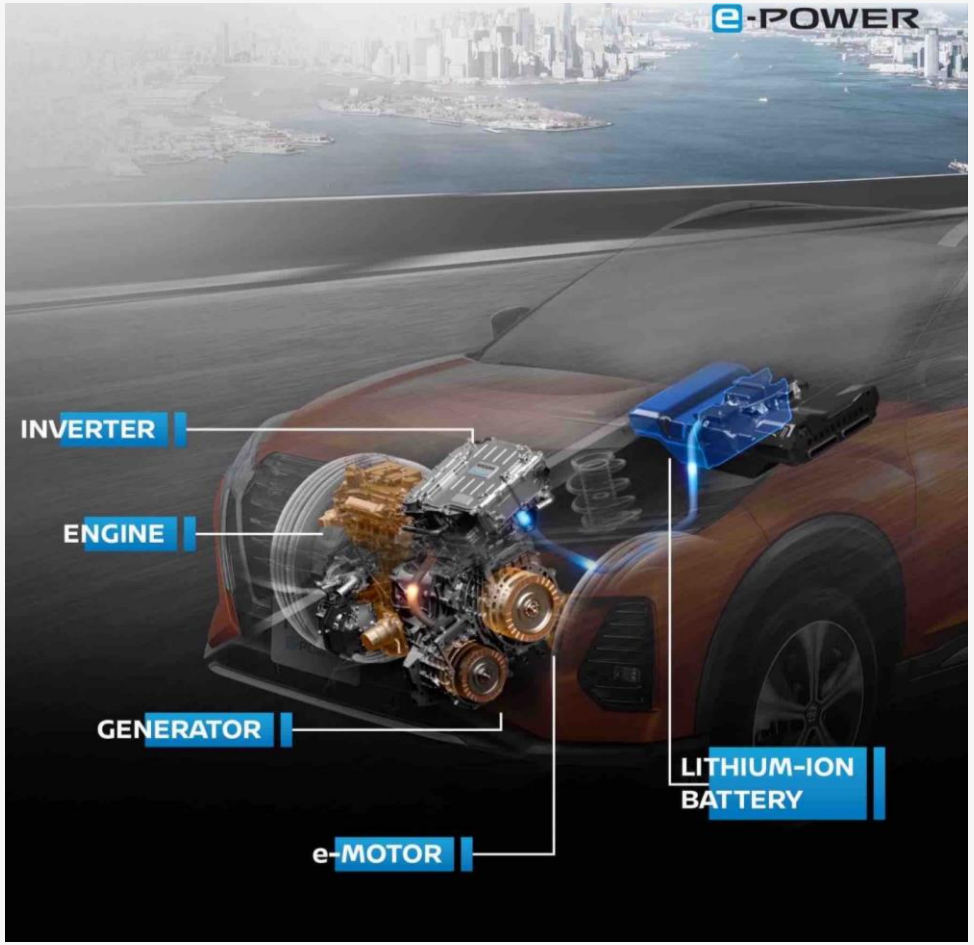
Introduction

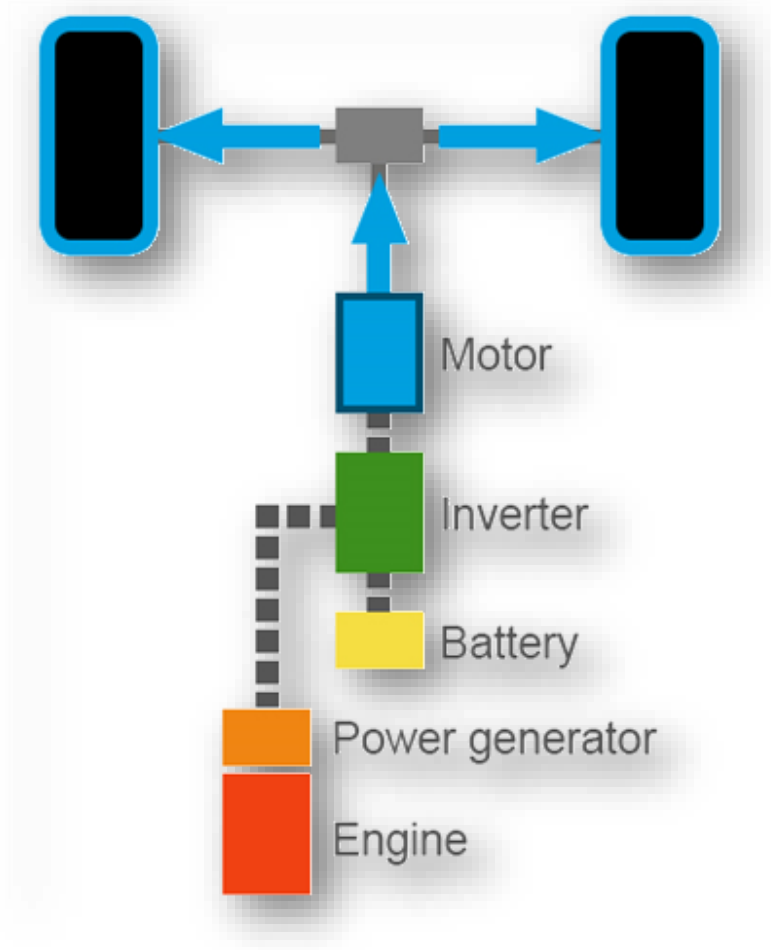


Currently OEM's are focusing more on vehicle electrification to reduce the emission and e-Power is one of the important pillar of electrification strategy. e-Power system is new form of eclectic motor-driven powertrain adopted by Nissan car in 2016.

The e-POWER system offers full electric motor drive, meaning that the wheels are completely driven by the electric motor. This system combine a electric motor with a gasoline engine that recharges the battery.

e-Power allows driver to enjoy the benefits of EV without worrying the battery charging.

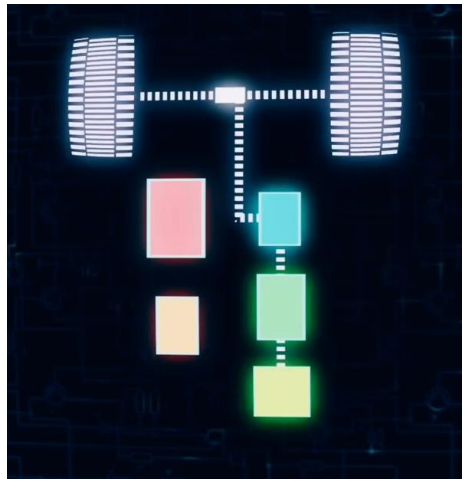
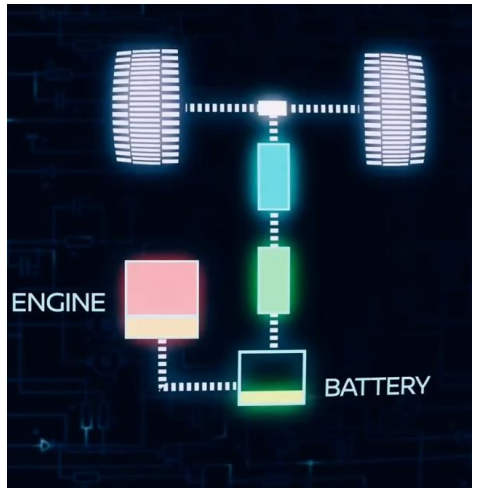
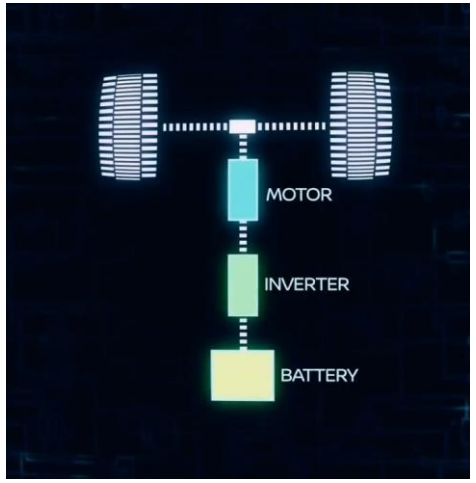
System Architecture



e-POWER system has high-output battery and the powertrain which is integrated with gasoline engine, power generator, inverter and a motor.

The electric motor provides power to the wheels. The inverter converts DC and AC to manage electric power between the battery and electric motor. Adjustments to the electric motor are made within fractions of a second for a responsive, refined driving experience. This vehicle uses a lightweight, compact, high-performance lithium-ion battery. The gasoline engine generates electricity to charge the battery, which powers the electric motor. The power source is the engine, rather than just the battery unlike a full EV.

System Comparison

Description	Hybrid	E-Power	EV
			
Driven by	Engine & Motor	Motor	Motor
Motor Specification	Low-Output Motor	High-Output Motor	High-Output Motor
Battery Capacity	NA	Around 1.41 KWH	Around 60KWH

Advantages

Massive Torque

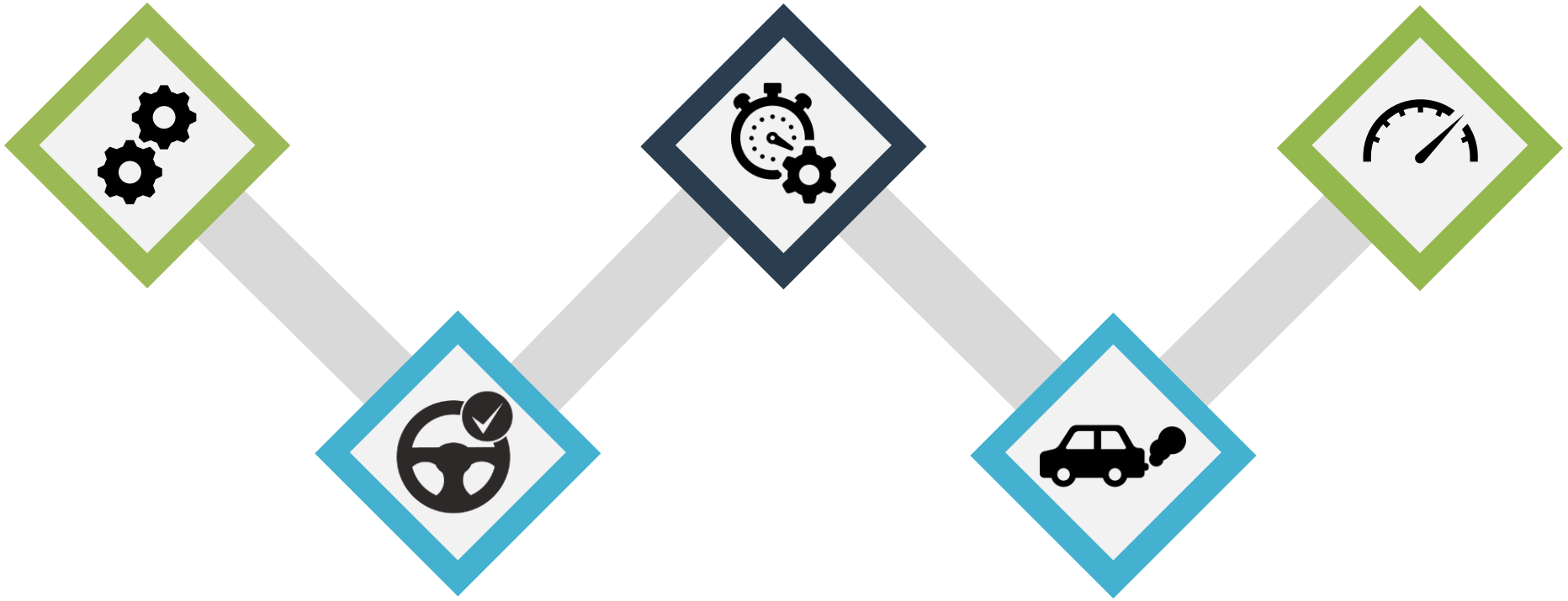
e-POWER delivers massive torque instantly, which enhances driving response & results in smooth acceleration

Max Thermal Efficiency

in e-POWER system the compression ratio is much higher than conventional engine this help to achieve Max Thermal Efficiency

Max Cruising Range

The e-POWER system allows you to enjoy all benefits of an EV without having to worry about charging the battery



Smooth & Quite Drive

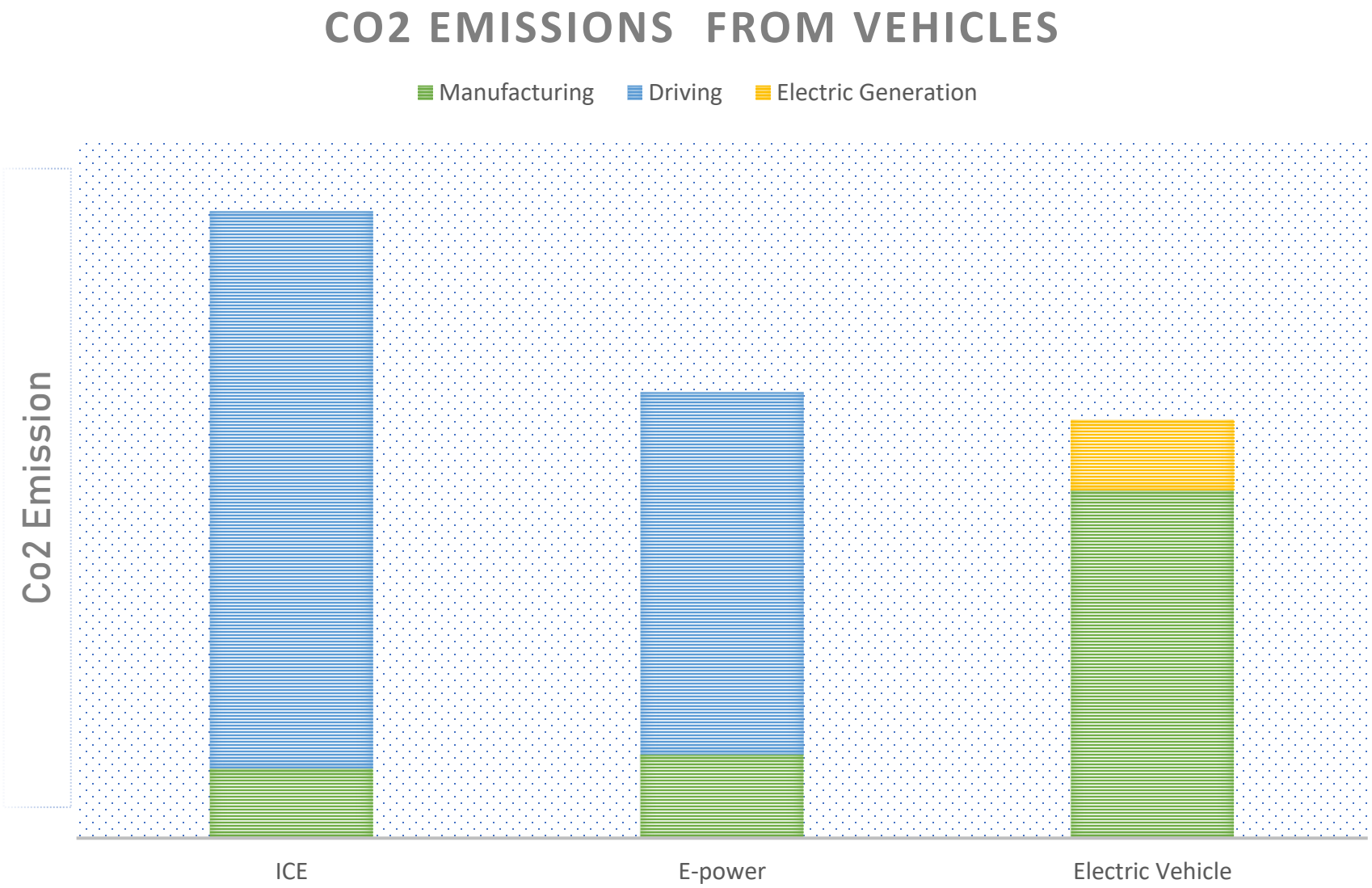
As e-POWER system uses Electric Motor powertrain so the driving experience is much smooth & Quite

Low Emission

Engine starts only when the battery need to charge and as there is no variation in load it works in its optimum state

"INDIAN SINERIO" Vehicle Charging Station and Road condition in India is still in developing phase so e-power will gave us time to built this infrastructure

Co2 Emission by vehicles

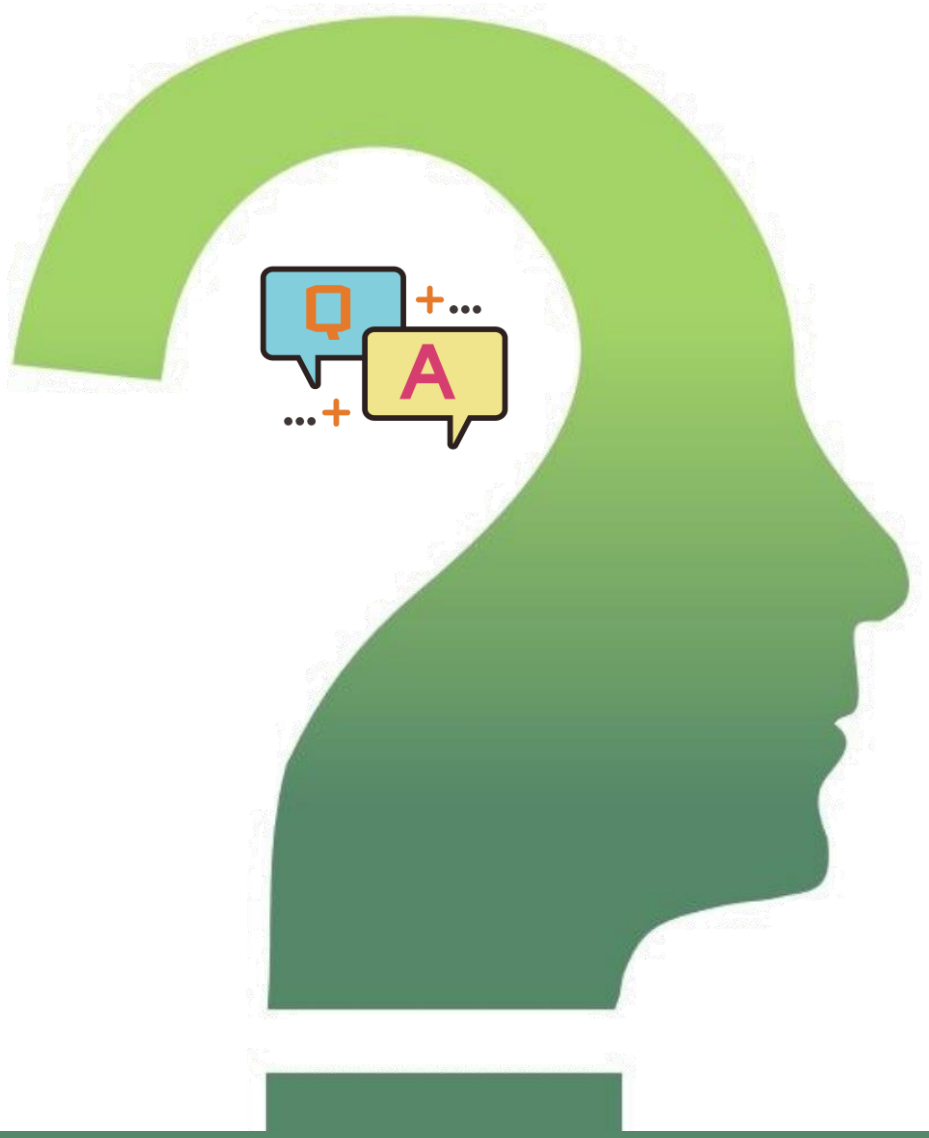


Benchmarking Data

Nissan Model	Note	Serena	Kicks	Qashqai	Sylphy
Body Type	Hatchback	Mini-van	SUV	SUV	Sedan
Launch Year	2016	2018	2020	2021	2021
Market	Japan	Japan	Japan	Japan + Euro	Japan + China
Cruising Range	1300 km	908 km	1107 km	950 km	900 km

Fuel Economy of e-power is 35 km/l

Open Discussion



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Always aim at perfection for only then will you achieve excellence

- J R D Tata

Thank You